

(a) Briefly explain why different ligands cause different absorption spectra in complexes of first-row transition metals. In the 'Spectrochemical Series' neutral molecules sometimes appear above anionic species. Account for this observation.

(b) Use the ideas of simple crystal field theory to identify the geometry of the four-coordinate complexes below from their magnetic behaviour:

$[\text{NiCl}_4]^{2-}$ paramagnetic

$[\text{Ni}(\text{CN})_4]^{2-}$ diamagnetic

Explain your reasoning clearly. What spin-only magnetic moment (measured in Bohr magnetons) would you expect for the paramagnetic complex?

a) Different ligands cause different absorption spectra in complexes of first-row transition metals as the identity of the ligand can cause the d-subshell to split to different extents. The ΔO between the two energy levels of the split d-subshell determines the wavelength of light absorbed; ΔO is proportional to the frequency of EMR absorbed.

little ΔO $\text{Cl} < \text{CN}$ big ΔO

$[\text{NiCl}_4]^{2-}$ paramagnetic: unpaired electrons
 $[\text{Ni}(\text{CN})_4]^{2-}$ diamagnetic: no unpaired electrons
 $\text{Ni}^{2+}: d^8$

Question 3
 Use the substitu

$\tan b = \frac{x}{z}$
 $b = \arctan$
 $\frac{dy}{dx} = \frac{1}{1+x^2}$
 $= \frac{1}{1+x^2}$
 $= \frac{1}{1+x^2}$

$\int \frac{1}{1+x^2} dx = \arctan x$
 $\tan^2 \theta = \frac{y^2}{x^2}$
 $\tan^2 \theta = \frac{y^2}{x^2}$
 $\int \frac{1}{4+t^2} dt = \frac{1}{4} \arctan \frac{t}{2}$
 $= \frac{1}{4} \arctan \frac{t}{2}$
 $\text{let } u = \frac{t}{2}$
 $\frac{du}{dt} = \frac{1}{2}$
 $dt = 2 du$
 $\int \frac{1}{4+t^2} dt = \int \frac{1}{4+4u^2} 2 du = \int \frac{1}{2(1+u^2)} du = \frac{1}{2} \arctan u$
 $= \frac{1}{2} \arctan \frac{t}{2}$

high spin
 low spin

$[\text{Ni}(\text{CN})_4]^{2-}$ is square planar
 $[\text{NiCl}_4]^{2-}$ is tetrahedral

$\mu = \sqrt{n(n+2)} \text{ BM}$
 $= 2.828$

$\mu = 2.828$